

Yixuan Wang

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PROFILE

Ph.D. Candidate and Researcher with experience in designing and deploying production ML systems, advancing human-AI interaction. Focus: LLMs, explainable AI, object detection, and model uncertainty quantification.

- Engineered advanced simulation frameworks and scalable data pipelines to bridge research and production.
- Developed rigorous evaluation protocols and designed trustworthy, explainable ML systems for AI reliability.
- Published research on XAI, uncertainty quantification, and human-AI trust in top-tier journals and conferences.

PROFESSIONAL EXPERIENCE

- Research Assistant (Ph.D.), Arizona State University** Jan. 2021 - Present
- Engineered scalable ML platforms with user interaction interface using Python, Pytorch and Flask, enabling deployment and real-time evaluation of models at scale, processing data from 200+ users for continuous performance improvement.
 - Enhanced human-AI interactions and decision-making with explainable AI (XAI) and model uncertainty quantification, resulting in publications in IJHCI and IEEE TVCG.
 - Developed simulation-based framework using Python and PyTorch to quantify cost-benefit tradeoffs in interactive machine learning systems for labeling with various machine learning models (e.g., BiLSTM, CNN, SGD), with findings published in IEEE TVCG.
 - Collaborated with cross-functional research team to design ML evaluation methodologies and experimental protocols for assessing model trustworthiness.
- Visiting Researcher, University of Illinois Urbana-Champaign** May 2023 - Aug. 2023
- Researched federated learning models for non-IID data distributions, implementing novel strategies to enhance privacy-preserving protocols and ensure algorithmic fairness across heterogeneous distributed systems.
 - Designed and developed a full-stack visual analytics system to assist experts in monitoring federated learning model performance, privacy metrics, and deployment outcomes.
- Data Engineer Intern, Wayfair** Jul. 2019 - Dec. 2019
- Engineered and maintained robust data pipelines and architectures using Python and SQL, ensuring data cube integrity and reliable analytics for the Promotion business unit.
 - Developed a full-stack monitoring system (Flask, React) with real-time alerts and front-end interface, enhancing visibility, automation, and efficiency across data engineering workflows.

SELECTED PUBLICATION

- **[IJHCI 2025] Yixuan Wang**, Jieqiong Zhao, Yang Ba, Michelle V Mancenido, Erin K Chiou, Ross Maciejewski. Impact of EXplainable AI on Trust Evolution with AI Error Severity: Comparing Similar Instances and Saliency Map in a Baggage Screening Task. [\[PDF\]](#)
- **[TVCG 2024] Yixuan Wang**, Jieqiong Zhao, Jiayi Hong, Ronald G Askin, Ross Maciejewski. A simulation-based approach for quantifying the impact of interactive label correction for machine learning. [\[PDF\]](#)
- **[JAIR 2024] Pouria Salehi**, Yang Ba, Nayoung Kim, Ahmadreza Mosallanezhad, Anna Pan, Myke C Cohen, **Yixuan Wang**, Jieqiong Zhao, Shawaiz Bhatti, James Sung, Erik Blasch, et al. Towards trustworthy AI-enabled decision support systems: Validation of the multisource AI scorecard table (MAST). [\[PDF\]](#)
- **[TVCG 2023] Jieqiong Zhao**, **Yixuan Wang**, Michelle V Mancenido, Erin K Chiou, Ross Maciejewski. Evaluating the impact of uncertainty visualization on model reliance. [\[PDF\]](#)

EDUCATION

Arizona State University, Ph.D. in Computer Science	2021 Jan. - 2026 May (Exp.)
Northeastern University, M.S. in Computer Systems Engineering: Software Design	2018 Sep. - 2020 Dec.
Central South University, B.E in Automation	2013 Sep. - 2017 May

SKILLS

Languages/Frameworks: Python, SQL, Java, JavaScript, PyTorch, TensorFlow, Git, JMP, Flask, React, Node.js
Research & Evaluation A/B Testing, Data Visualization, Statistical Modeling, User Studies (Qualtrics, MTurk, Prolific)